Digital Signal Processing (DSP) - Session 4

The Z-Transform and Its Application to the Analysis of LTI Systems

Example 1.2

Determine the z-transform of the signal

$$x(n) = (\frac{1}{2})^n u(n)$$

Solution

1. **Apply the Z-Transform Definition**: The z-transform is defined by the power series:
$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$
2. **Substitute the Signal**: Since the unit step signal $u(n)$ is 1 for $n \geq 0$ and 0 for $n < 0$, change the summation limits:
$$X(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n}$$
3. **Rewrite as a Single Power**: Combine the terms into a geometric series form:
$$X(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2} z^{-1}\right)^n$$
4. **Use Geometric Series Summation**: Recall the sum formula for an infinite geometric series: $\sum_{n=0}^{\infty} A^n = \frac{1}{1-A}$ if $|A| < 1$. Here, $A = \frac{1}{2}z^{-1}$.
5. **Determine the Closed-Form Expression**: Substitute A into the formula:
$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}$$
6. **Find the Region of Convergence (ROC)**: The series converges only if $|\frac{1}{2}z^{-1}| < 1$. Solving for z: $\frac{1}{2}|z|^{-1} < 1 \implies \frac{1}{2|z|} < 1 \implies |z| > \frac{1}{2}$.

Example 1.3

Determine the **z-transform** of the exponential signal:

$$x(n) = \alpha^n u(n) = \begin{cases} \alpha^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

Solution

1. **Use the Z-Transform Definition**: Apply the standard power series formula:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

2. **Substitute the Signal**: Since the unit step $u(n)$ is zero for $n < 0$, adjust the summation limits:

$$X(z) = \sum_{n=0}^{\infty} \alpha^n z^{-n}$$

3. **Rewrite as a Geometric Series**: Combine the terms into a single power of n:

$$X(z) = \sum_{n=0}^{\infty} (\alpha z^{-1})^n$$

4. **Apply Geometric Series Summation**: Use the sum formula $\sum_{n=0}^{\infty} A^n = \frac{1}{1-A}$, which is valid for $|A| < 1$.
5. **Form the Closed-Form Expression**: Substitute $A = \alpha z^{-1}$ into the sum formula:

$$X(z) = \frac{1}{1 - \alpha z^{-1}}$$

6. **Determine the Region of Convergence (ROC)**: The series converges only if $|\alpha z^{-1}| < 1$. Solving for z yields: $|z| > |\alpha|$

Example 1.4

Determine the **z-transform** of the anticausal signal:

$$x(n) = -\alpha^n u(-n-1) = \begin{cases} 0, & n \geq 0 \\ -\alpha^n, & n \leq -1 \end{cases}$$

Solution

1. **Use the Z-Transform Definition**: Apply the standard power series formula: $X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$.
2. **Substitute the Signal**: Adjust the limits for the anticausal signal:

$$X(z) = \sum_{n=-\infty}^{-1} (-\alpha^n)z^{-n}$$

3. **Change Variable for Summation**: Let $l = -n$. The sum becomes:

$$X(z) = -\sum_{l=1}^{\infty} (\alpha^{-1}z)^l$$

4. **Apply Geometric Series Summation**: Use the formula $A + A^2 + A^3 + \cdots = \frac{A}{1-A}$ for $|A| < 1$.

5. **Form the Closed-Form Expression**: Substitute $A = \alpha^{-1}z$ into the formula and simplify:
$$X(z) = -\frac{\alpha^{-1}z}{1-\alpha^{-1}z} = \mathbf{\frac{1}{1 - \square z^{-1}}}$$
6. **Determine the Region of Convergence (ROC)**: The series converges only if $|\alpha^{-1}z| < 1$. Solving for z yields the interior of a circle: $|z| < |\alpha|$

Note: Examples 1.3 and 1.4 highlight that a closed-form expression does not uniquely specify a signal; both the expression and the **ROC** are required for uniqueness.

Example 1.5

Determine the **z-transform** of the two-sided signal:

$$x(n) = \alpha^n u(n) + b^n u(-n-1)$$

Solution

1. **Use the Z-Transform Definition**: Apply the standard power series formula: $X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$.
2. **Substitute the Signal Components**: Split the summation into two parts corresponding to the causal and anticausal terms:

$$X(z) = \sum_{n=0}^{\infty} \alpha^n z^{-n} + \sum_{n=-\infty}^{-1} b^n z^{-n}$$

3. **Rewrite as Geometric Series**:
   - **Part 1**: $\sum_{n=0}^{\infty} (\alpha z^{-1})^n$.
   - **Part 2**: Let $l = -n$. The sum becomes $\sum_{l=1}^{\infty} (b^{-1}z)^l$.
4. **Determine Convergence Conditions**:
   - The first series converges if $|\alpha z^{-1}| < 1 \implies |z| > |\alpha|$.
   - The second series converges if $|b^{-1}z| < 1 \implies |z| < |b|$.
5. **Analyze the Region of Convergence (ROC)**:
   - If $|b| < |\alpha|$, there is no common region of convergence; thus, $X(z)$ **does not exist**.
   - If $|\alpha| < |b|$, the common region is the ring between the two circles: $|\alpha| < |z| < |b|$.
6. **Form the Closed-Form Expression**: If the ROC exists ($|\alpha| < |b|$), combine the results of the two geometric series:

$$X(z) = \frac{1}{1-\alpha z^{-1}} - \frac{1}{1-bz^{-1}}$$

   Which simplifies to:

$$X(z) = \frac{(b-\alpha)z^{-1}}{(1-\alpha z^{-1})(1-bz^{-1})}$$

The Inverse z-Transform

- **Definition**: The procedure used to recover the time-domain signal $x(n)$ from its complex-plane representation $X(z)$.
- **Mathematical Basis**: Derived using the **Cauchy integral theorem**.
- **Inversion Formula**:
$$x(n) = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz$$
where C is a counterclockwise closed contour within the **Region of Convergence (ROC)** that encloses the origin.
- **Practical Methods of Inversion**:
  1. **Contour Integration**: Direct evaluation of the integral by summing the residues of $X(z)z^{n-1}$ at the poles inside the contour C.
  2. **Power Series Expansion**: Expanding $X(z)$ into a series (often via **long division**); the coefficients of the resulting z^{-n} terms represent the values of $x(n)$.
  3. **Partial-Fraction Expansion**: Decomposing a rational z-transform into simpler fractions that can be inverted using a **table lookup** and the linearity property.

Properties of the z-Transform

- **Linearity**: The z-transform of a linear combination of signals equals the same linear combination of their individual transforms:
$$a_1 x_1(n) + a_2 x_2(n) \leftrightarrow a_1 X_1(z) + a_2 X_2(z)$$
- **Time shifting**: Shifting a signal by k samples corresponds to multiplication by z^{-k}:
$$x(n-k) \leftrightarrow z^{-k} X(z)$$
The ROC remains unchanged except possibly at $z = 0$ or $z = \infty$.
- **Scaling in the z-domain**: Multiplying a signal by an exponential sequence a^n scales the complex variable z:
$$a^n x(n) \leftrightarrow X(a^{-1} z)$$
The ROC is scaled by $|a|$.
- **Time reversal**: Reflecting a signal in the time domain corresponds to inverting the z variable:
$$x(-n) \leftrightarrow X(z^{-1})$$
The ROC is inverted $(1/r_2 < |z| < 1/r_1)$.
- **Differentiation in the z-domain**: Multiplying a signal by n corresponds to differentiation in the z-domain:
$$n x(n) \leftrightarrow -z \frac{dX(z)}{dz}$$
The ROC remains the same.
- **Convolution of two sequences**: Convolution in the time domain is equivalent to multiplication in the z-domain:
$$x_1(n) * x_2(n) \leftrightarrow X_1(z) X_2(z)$$
The ROC is at least the intersection of individual ROCs.

- **Correlation of two sequences**: The transform of the crosscorrelation of two signals is:

$$r_{x_1x_2}(l) \leftrightarrow R_{x_1x_2}(z) = X_1(z)X_2(z^{-1})$$

- **Multiplication of two sequences**: Multiplication in the time domain corresponds to a complex convolution integral in the z-domain:

$$x_1(n)x_2(n) \leftrightarrow \frac{1}{2\pi j}\oint_C X_1(v)X_2(z/v)v^{-1}dv$$

- **Parseval's relation**: Relates the energy of signals in the time domain to an integral in the z-domain:

$$\sum_{n=-\infty}^{\infty} x_1(n)x_2^*(n) = \frac{1}{2\pi j}\oint_C X_1(v)X_2^*(1/v^*)v^{-1}dv$$

- **The Initial Value Theorem**: For a causal signal ($x(n) = 0$ for $n < 0$), the initial value can be found by:

$$x(0) = \lim_{z\to\infty} X(z)$$

Example 2.1

Determine the **z-transform** and the **Region of Convergence (ROC)** of the signal:

$$x(n) = [3(2^n) - 4(3^n)]u(n)$$

Solution

1. **Define Signal Components**: Let $x_1(n) = 2^n u(n)$ and $x_2(n) = 3^n u(n)$.
2. **Apply Linearity**: The signal is a linear combination: $x(n) = 3x_1(n) - 4x_2(n)$. The property states: $X(z) = 3X_1(z) - 4X_2(z)$.
3. **Find Individual Transforms and ROCs**:
   - **For** $x_1(n)$: Using the pair $\alpha^n u(n) \leftrightarrow \frac{1}{1-\alpha z^{-1}}$, we get $X_1(z) = \frac{1}{1-2z^{-1}}$ with **ROC:** $|z| > 2$.
   - **For** $x_2(n)$: Similarly, $X_2(z) = \frac{1}{1-3z^{-1}}$ with **ROC:** $|z| > 3$.
4. **Form the Combined Expression**: Substitute these into the linear equation:

$$X(z) = \frac{3}{1-2z^{-1}} - \frac{4}{1-3z^{-1}}$$

5. **Determine Overall ROC**: The overall ROC is the **intersection** of the individual ROCs ($|z| > 2$ and $|z| > 3$). **ROC:** $|z| > 3$

Proof of the time shifting property

Let $y(n) = x(n-k)$ be the shifted signal. We determine its z-transform $Y(z)$ using the definition:

$$Y(z) = \sum_{n=-\infty}^{\infty} y(n)z^{-n} = \sum_{n=-\infty}^{\infty} x(n-k)z^{-n}$$

To solve this, we make a **change of variable**. Let $m = n - k$, which implies that $n = m + k$. As n ranges from $-\infty$ to ∞, m also ranges from $-\infty$ to ∞. Substituting these into the summation:

$$Y(z) = \sum_{m=-\infty}^{\infty} x(m)z^{-(m+k)}$$

Using the properties of exponents ($z^{-(m+k)} = z^{-m}z^{-k}$), we can rewrite the expression:

$$Y(z) = \sum_{m=-\infty}^{\infty} x(m)z^{-m}z^{-k}$$

Since z^{-k} does not depend on the summation index m, we can factor it out of the sum:

$$Y(z) = z^{-k} \sum_{m=-\infty}^{\infty} x(m)z^{-m}$$

We recognize that the remaining summation is the definition of the z-transform $X(z)$. Therefore:

$$Y(z) = z^{-k}X(z)$$

Proof of the convolution of two sequences property

The convolution of $x_1(n)$ and $x_2(n)$ is defined as:

$$x(n) = x_1(n) * x_2(n) = \sum_{k=-\infty}^{\infty} x_1(k)x_2(n-k)$$

By the definition of the direct z-transform, we have:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} = \sum_{n=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} x_1(k)x_2(n-k) \right] z^{-n}$$

Rearrange the terms to group those dependent on n:

$$X(z) = \sum_{k=-\infty}^{\infty} x_1(k) \left[\sum_{n=-\infty}^{\infty} x_2(n-k)z^{-n} \right]$$

The inner summation is the z-transform of the shifted sequence $x_2(n-k)$. According to the **time-shifting property**,

$$x_2(n-k) \leftrightarrow z^{-k}X_2(z)$$

Substituting this into the equation:

$$X(z) = \sum_{k=-\infty}^{\infty} x_1(k) \left[z^{-k}X_2(z) \right]$$

Since $X_2(z)$ does not depend on the summation index k, it can be moved outside the sum:

$$X(z) = X_2(z) \sum_{k=-\infty}^{\infty} x_1(k)z^{-k}$$

The remaining sum is the definition of $X_1(z)$. Thus:

$$X(z) = X_1(z)X_2(z)$$

Poles and Zeros

- **Definitions**:
  - **Zeros**: The values of z for which $H(z) = 0$.
  - **Poles**: The values of z for which $H(z) = \infty$.
- **Rational System Function (Standard Form)**: Assuming $a_0 = 1$ for a linear constant-coefficient difference equation:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

- **Factored (Pole-Zero) Form**: Factoring the polynomials reveals the specific locations of the zeros (z_k) and poles (p_k):

$$H(z) = Gz^{N-M} \frac{(z - z_1)(z - z_2) \dots (z - z_M)}{(z - p_1)(z - p_2) \dots (z - p_N)}$$

  - **Gain Factor** (G): Defined as b_0/a_0.
  - **Trivial Singularities**: The term z^{N-M} represents additional zeros (if $N > M$) or poles (if $N < M$) located at the **origin** ($z = 0$).
- **Graphical Representation (Pole-Zero Plot)**:
  - **Poles** are indicated by crosses ($\times$).
  - **Zeros** are indicated by circles ($\circ$).
  - **Multiplicity**: If a pole or zero occurs more than once at the same location, the order is noted next to the mark.
- **Relationship to ROC**:
  - By definition, the **Region of Convergence (ROC)** of a z-transform cannot contain any poles.
  - The transform does not converge at a pole because the value becomes infinite.

Example

Determine $H(z)$ for this system:

$$y(n) = x(n) - \frac{1}{2}x(n-2) - y(n-1)$$

Solution

1. Rearrange the Difference Equation

First, group all the terms involving the output $y(n)$ on one side and the terms involving the input $x(n)$ on the other:

$$y(n) + y(n-1) = x(n) - \frac{1}{2}x(n-2)$$

2. Apply the Z-Transform

Next, take the z-transform of both sides of the equation. According to the **time-shifting property**, a delay of k samples in the time domain corresponds to multiplying by z^{-k} in the z-domain ($x(n-k) \leftrightarrow z^{-k}X(z)$):

$$Y(z) + z^{-1}Y(z) = X(z) - \frac{1}{2}z^{-2}X(z)$$

3. Factor and Solve for $H(z)$

Factor out $Y(z)$ and $X(z)$ from their respective sides:

$$Y(z)(1 + z^{-1}) = X(z)\left(1 - \frac{1}{2}z^{-2}\right)$$

The system function $H(z)$ is defined as the ratio of the output transform to the input transform:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - \frac{1}{2}z^{-2}}{1 + z^{-1}}$$

Example 3.1

Determine the pole-zero plot for the signal

$$x(n) = a^n u(n), \quad a > 0$$

Solution

1. **Compute the z-transform**: Using the direct z-transform formula,

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n$$

   This geometric series converges when $|az^{-1}| < 1$, resulting in:

$$X(z) = \frac{1}{1 - az^{-1}}$$

2. **Express as a rational function**: Convert the expression to use positive powers of z by multiplying the numerator and denominator by z:

$$X(z) = \frac{z}{z - a}$$

3. **Identify zeros and poles**:
   - **Zeros**: The values of z where the numerator is zero. Here, $X(z) = 0$ at $z_1 = 0$.
   - **Poles**: The values of z where the denominator is zero (making $X(z)$ infinite). here, $X(z) = \infty$ at $p_1 = a$

4. **Determine the Region of Convergence (ROC)**: For a causal signal, the ROC is the exterior of a circle of radius equal to the magnitude of the largest pole. Thus, the **ROC is** $|z| > a$

Example 3.2

Determine the pole-zero plot for the signal:

$$x(n) = \begin{cases} a^n, & 0 \le n \le M-1 \\ 0, & \text{elsewhere} \end{cases}$$

where $a > 0$.

Solution

1. **Apply the Z-transform definition:**

$$X(z) = \sum_{n=0}^{M-1} a^n z^{-n} = \sum_{n=0}^{M-1} (az^{-1})^n$$

2. **Use the finite geometric series formula:**

$$X(z) = \frac{1-(az^{-1})^M}{1-az^{-1}}$$

3. **Rearrange into positive powers of z:** Multiply the numerator and denominator by z^M:

$$X(z) = \frac{z^M - a^M}{z^{M-1}(z-a)}$$

4. **Find the zeros:** Zeros occur where the numerator is zero: $z^M - a^M = 0$. The M roots are: $z_k = ae^{j2\pi k/M}$ for $k = 0, 1, \dots, M-1$.
5. **Find the poles:** The denominator indicates $M-1$ poles at $z = 0$ (the origin) and one pole at $z = a$.
6. **Identify pole-zero cancellation:** For $k = 0$, the zero is $z_0 = ae^{j0} = a$. This **zero at $z = a$ cancels the pole at $z = a$.**
7. **Final Result:**
   - **Zeros:** $M-1$ zeros located on a circle of radius a at $z_k = ae^{j2\pi k/M}$ for $k = 1, 2, \dots, M-1$.
   - **Poles:** $M-1$ poles located at the origin ($z = 0$).

Analysis of Linear Time-Invariant Systems in the z-Domain

- **Response of Systems with Rational System Functions**
  - Linear time-invariant (LTI) systems described by constant-coefficient difference equations have **rational system functions** of the form

$$H(z) = \frac{B(z)}{A(z)}$$

- For a relaxed system, the output z-transform is the product of the system function and the input z-transform:

$$Y(z) = H(z)X(z)$$

- The total response $y(n)$ can be subdivided into two parts:
  * **Natural Response**: Determined by the poles p_k of the system.
  * **Forced Response**: Determined by the poles q_k of the input signal.
- The system's influence on the forced response and the input signal's influence on the natural response are exerted through scale factors (residues) determined during partial-fraction expansion.

- **Transient and Steady-State Responses**
  - **Transient Response**: In a stable system, all poles lie inside the unit circle ($|p_k| < 1$), causing the natural response to decay to zero as $n \to \infty$.
  - **Steady-State Response**: If the input signal is a sinusoid (poles on the unit circle), the forced response persists for all $n \geq 0$.
  - A system only sustains a steady-state output if the input signal itself persists (e.g., a sinusoid applied at $n = 0$ that does not decay).

- **Causality and Stability**
  - **Causality**: An LTI system is causal if and only if the region of convergence (ROC) of its system function is the **exterior of a circle**, including $z = \infty$.
  - **Stability**: An LTI system is BIBO stable if and only if the **unit circle** ($|z| = 1$) **is contained within the ROC** of the system function $H(z)$.
  - **Combined Requirement**: A causal LTI system is BIBO stable if and only if **all poles of** $H(z)$ **lie strictly inside the unit circle**.

- **Pole-Zero Cancellations**
  - Occurs when a pole and zero are at the identical location, causing the term associated with that pole to vanish from the inverse z-transform.
  - This can be used to **reduce the order of a system** or to suppress specific modes of an input signal.
  - In practice, one should not attempt to stabilize an unstable system by placing a zero over an unstable pole because any numerical imprecision will prevent exact cancellation, likely leaving the system unstable.

- **Multiple-Order Poles and Stability**
  - If a system pole lies on the unit circle, the system is generally **unstable** because a matching pole in a bounded input (like a unit step) will create a multiple-order pole on the unit circle, resulting in an unbounded output (e.g., a ramp signal).
  - Stable systems must have poles strictly inside the unit circle; matching input poles will still result in terms like $n^b p_k^n$ in the output, but these will decay as $n \to \infty$ because the exponential factor p_k^n dominates the polynomial n^b when $|p_k| < 1$.
  - Systems with distinct poles on the unit circle (like digital oscillators) are often called **marginally stable**.

- **Stability of Second-Order Systems**

- A causal two-pole system with

$$H(z) = \frac{b_0}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

  is stable if its poles lie inside the unit circle.
- This condition is satisfied if the coefficients (a_1, a_2) fall within the **stability triangle** defined by:
  * $|a_2| < 1$
  * $|a_1| < 1 + a_2$
- The unit sample response $h(n)$ behavior changes based on where the coefficients lie in this triangle: real and distinct poles, real and equal poles, or complex-conjugate poles.

Digital Signal Processing in Python

ECG Signal Analysis

Loading the ECG Data

```
from scipy.datasets import electrocardiogram
ecg = electrocardiogram()
```

Loads a real-world ECG (electrocardiogram) sample dataset from SciPy. It's a 1D NumPy array of heart voltage readings.

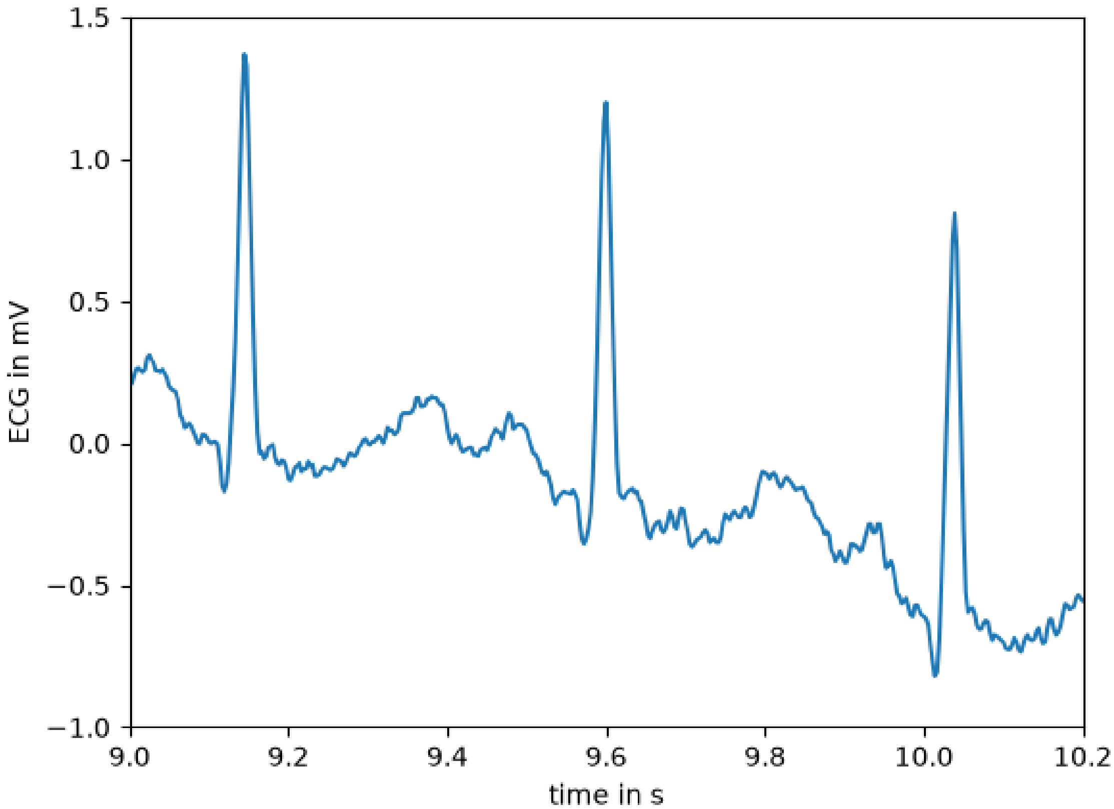
Inspecting the Data

```
# confirms it's a NumPy array
type(ecg)
# shape, mean, and standard deviation
ecg.shape, ecg.mean(), ecg.std()
# total number of samples
ecg.size
```

Plotting ECG Segments

```
import numpy as np
import matplotlib.pyplot as plt

fs = 360
time = np.arange(ecg.size) / fs
plt.plot(time, ecg)
plt.xlabel("time in s")
plt.ylabel("ECG in mV")
plt.xlim(9, 10.2)
plt.ylim(-1, 1.5)
plt.show()
```



The signal was recorded at **fs = 360 Hz**, so `time = np.arange(ecg.size) / fs` converts sample indices to seconds.

Audio Recording & Playback

Recording Audio

```
import sounddevice
from scipy.io.wavfile import write
fs= 44100
second = int(input("enter time duration in second:"))
print("recording...")
record_voice= sounddevice.rec(int(second * fs), samplerate=fs, channels=1)
sounddevice.wait()
write("output.wav", fs, record_voice)
print("recording complete")
```

Records audio from the microphone for a user-specified duration at 44100 Hz and saves it as `output.wav`.

Listing Audio Devices

```
sounddevice.query_devices()
```

Lists all available audio input/output devices on the system.

Playback

Two ways to play back the saved file:

1. **sounddevice + soundfile** — reads the WAV and plays it via the system audio device

```
import sounddevice
import soundfile as sf

data, fs = sf.read("output.wav", dtype="float32")
sounddevice.play(data, fs)
sounddevice.wait()
```

2. **IPython.display.Audio** — embeds an interactive audio player directly in the notebook

```
from IPython.display import Audio
Audio("output.wav")
```

Visualizing the Audio Waveform

```
fs = 44100
time = np.arange(data.size) / fs
plt.plot(time, data)
plt.xlabel("time in s")
plt.ylabel("amplitude")
plt.xlim(0.5, 0.6)
plt.ylim(-1.5, 1.5)
plt.show()
```

Plots the amplitude of the audio signal over time, zoomed into a 100ms window (0.5–0.6s).

Complex Numbers (Review/Intro)

Complex Square Root

```
cmath.sqrt(-1)  # → 1j
```

Defining & Inspecting a Complex Number

```
z = 2 + 3j
np.real(z)   # 2.0
np.imag(z)   # 3.0
np.abs(z)    # magnitude
np.angle(z)  # phase angle in radians
```

Visualizations

- **Cartesian plot** — plots z as a point on the real/imaginary plane with axes drawn

```
# plot the complex number
plt.plot(np.real(z),np.imag(z), 'ks')
```

```
# make plot look nicer
plt.xlim([-5,5])
plt.ylim([-5,5])
plt.plot([-5,5], [0,0],'k')
plt.plot([0,0], [-5,5],'k')
plt.xlabel('real axis')
plt.ylabel('imag axis')
plt.show()
```

- **Polar plot** — uses `plt.polar()` to draw a vector from the origin with its magnitude and angle

```
mag = np.abs(z)
ang = np.angle(z)

plt.polar([0, ang], [0, mag], 'r')
plt.show()
```

Polynomial Roots

```
p = [1, 0, 0, 1]
np.roots(p)
```

Finds solutions to $x^3 + 1 = 0$.